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RESEARCH ARTICLE

SPAF: Enhancing YOLO-Based Object Detection Using Sparse Attention and Progressive Adaptive Fusion

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ABSTRACT Detecting objects in Autonomous Aerial Vehicles (AAVs) imagery remains a challenging task due to large variations in object sizes and complex backgrounds, which often leads to missed detections and reduced accuracy, especially for small objects that contain limited visual information and are easily lost during feature extraction. To address these challenges, this paper proposes SPAF, an enhanced YOLO-based model built upon YOLO26 that improves object detection through the integration of a multi-scale attention mechanism, progressive feature aggregation, and adaptive feature fusion. The proposed architecture enhances the backbone network by integrating a Sparse Vision Attention (SVA) module to facilitate the extraction of fine-grained and task-relevant features in complex aerial scenes. In the neck, the Progressive Bidirectional Auxiliary Pyramid Network (PBAP-Net) facilitates efficient cross-scale feature aggregation by combining top-down semantic information with bottom-up spatial details, thereby strengthening multi-scale representation. Finally, Adaptive Channel Multi-Scale Fusion Head(ACMF-Head) refines the final feature representations by adaptively fusing multi-scale information and reducing inconsistencies across feature levels, resulting in more reliable and robust predictions. Experimental results on the VisDrone2019 dataset demonstrate that SPAF achieves 46.8% mAP@0.5, which is 7.5% higher compared to the baseline model. These results confirm the effectiveness of the proposed approach for robust AAV object detection in challenging aerial environments.

INDEX TERMS AAV, attention, multi-scale feature fusion, object detection.

I. INTRODUCTION

The proliferation of AAVs has led to a transformation of various domains such as search and rescue, urban monitoring, and environmental surveillance [1]. This rapid expansion has significantly increased the demand for accurate and efficient automated object detection systems. In general, object detection is a fundamental computer vision task focused on identifying and localizing objects within images [2]. Accordingly, mainstream object detection algorithms can be divided into two types: traditional and deep learning. Early methods relied on handcrafted feature extraction using algorithms such as Canny, Sobel, and HOG [3]; however, they were limited in handling complex environments and required significant manual effort. With the emergence of deep learning, object detection evolved into a data-driven, end-to-end task, where Convolutional Neural Networks (CNNs) [4] automatically learn hierarchical features from raw images, improving both accuracy and efficiency. Nevertheless, CNN-based architectures are inherently limited due to local receptive fields, which restrict their ability to capture complex spatial relationships required for detecting small objects in large-scale scenes. However, object detection models can be generally classified into two-stage [2] and

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one-stage [5] approaches. Two-stage detectors, such as R-CNN and Faster R-CNN [6], first generate region proposals and then perform classification, achieving high accuracy but at the cost of increased computational complexity [2]. In contrast, one-stage detectors like YOLO (You Only Look Once) and SSD [7] perform localization and classification in a single pass, enabling faster real-time performance while maintaining competitive accuracy [2], [5]. Despite these advancements, object detection in aerial imagery has several challenges. First, the high altitude of AAVs results in extremely small objects that lack sufficient semantic information, making them prone to being lost during deep feature extraction [5]. Second, significant scale variation occurs as object sizes change with AAV altitude, necessitating adaptive multi-scale fusion strategies [5]. Third, cluttered backgrounds, occlusions, and motion blur further increase detection difficulty, often leading to high false-positive rates [5]. In addition to these challenges, data scarcity remains a major limitation in AAV-based object detection, where obtaining large-scale and accurately annotated datasets is often difficult and expensive. As highlighted by Alzubaidi et al. [8], deep learning models typically require large-scale labeled data to achieve robust generalization. However, in many real-world applications, including aerial imagery analysis, collecting sufficient annotated data is challenging due to the high cost, time-consuming nature of manual labeling, and the need for domain-specific expertise. Consequently, this limitation often leads to overfitting and reduced generalization performance, particularly for small object instances in complex AAV scenes. To address these challenges, we chose YOLO as the base model due to its exceptional speed and accuracy in real-time applications [9]. Despite recent advances, maintaining high detection accuracy in object detection without sacrificing computational efficiency remains an open challenge. To overcome this limitation, this paper proposes an enhanced multi-scale object detection framework based on the YOLO26 architecture. Our contributions are summarized as follows:

- We propose SPAF, an object detection model based on the YOLO26 architecture, specifically designed to address the challenges of small object detection and scale inconsistency in AAV-captured imagery.
- We leverage three key modules: Sparse Vision Attention (SVA), which utilizes sparse attention mechanisms to suppress redundant semantic information while enhancing high-order non-semantic feature representation; Progressive Bidirectional Auxiliary Pyramid Network (PBAP-Net), which enables effective multi-scale feature fusion by preserving low-level spatial details and integrating them with high-level semantics; and Adaptive Channel Multi-Scale Fusion Head (ACMF-Head), which dynamically maintains feature consistency and mitigates conflicts across different scales.
- Extensive experiments on the VisDrone2019 dataset demonstrate that SPAF achieves competitive object detection performance, with improved detection accuracy compared to the baseline model.

The remainder of this paper is organized as follows. Section II reviews related work on object detection and advanced attention mechanisms. Section III presents the proposed SPAF model in detail, describing the SVA module, the PBAP-Net, and the ACMF-Head. Section IV describes the experimental setup, dataset details, and evaluation protocols. Finally, Section V concludes the paper, summarizes the main findings, and discusses potential directions for future research.

II. RELATED WORK

This section provides a comprehensive review of recent advancements in object detection, focusing on state-of-the-art techniques for feature enhancement and multi-scale fusion.

A. MULTI-SCALE FEATURE ENHANCEMENT METHODS

Multi-scale feature enhancement plays a pivotal role in mitigating the loss of semantic information for tiny targets in aerial imagery. The SMA-YOLO model [10] proposes a multi-scale detection algorithm for AAV remote sensing, integrating NSSA attention into the backbone, BIMA-FPN as the neck, and a CSFA-Head to improve multi-scale fusion, achieving 45.9% mAP on the VisDrone2019 dataset and significantly enhancing small-object detection. Building upon this, YOLO-DroneMS [11] introduces a multi-scale detection network for AAV images, where the neck combines LSKA, SPPF, and ASF-DySample for adaptive upsampling, achieving a 3.6% mAP improvement over YOLOv8-n on VisDrone2019. To address the need for efficiency, LMSOE-Net [12] presents a lightweight design using an EMSP neck and FPSC, reducing parameters by 30% while slightly improving mAP_{50} by 1–2% on VisDrone2019 and DOTA-v1.5. Similarly, MGDFIS [13] combines deep semantic and shallow spatial features with dual-branch fusion and adaptive weighting, improving mAP by +3.5% while maintaining 72 FPS real-time speed. Furthermore, Dynamic-Tiling [14] proposes a model-agnostic inference strategy that divides images into adaptive overlapping tiles to improve small-object detection, while SO-DETR [15] combines spatial and frequency-domain features within a teacher–student framework to enhance small-object representation on VisDrone-2019-DET and AAVVaste benchmarks. Furthermore, QueryDet [16] introduces a cascaded sparse query mechanism that selectively activates high-resolution feature regions for small object detection, thereby reducing redundant computation while preserving detection accuracy in dense aerial scenes. Similarly, CEASC [17] proposes a context-enhanced adaptive sparse convolution framework that combines sparse feature extraction with global contextual modeling to improve detection robustness and computational efficiency for AAV-based object detection tasks. In transformer based approaches, DNTR [18] integrates denoising feature pyramids with transformer-based regional feature interaction to strengthen local global feature representation

and enhance tiny object detection performance in aerial imagery.

B. ATTENTION-BASED METHODS

Attention mechanisms have been widely adopted to capture long-range dependencies and strengthen inter-scale feature communication. CSSDet [19] employs attention re-weighting to achieve an mAP50 − 95 of 35.3%, demonstrating higher recall for small targets. Complementing this, AAV-DETR [20] adopts a hierarchical attention transformer to capture spatial dependencies, achieving improvements of +3.1% AP and +4.2% AP50 on VisDrone2019.

In terms of scale-adaptive architectural designs, SED-YOLO [21] builds upon YOLOv5s to tackle remote sensing challenges by utilizing Switchable Atrous Convolution to enlarge the field of view without losing fine details, alongside a lightweight attention mechanism to re-weight cross-scale features, yielding a +2.4% mAP boost on the DOTA benchmark.

C. OTHER METHODS

Beyond standard feature enhancement, other specialized methods have been explored to tackle extreme detection conditions. The approach utilizing adversarial learning for scale-invariant feature disentangling [22] separates scale-dependent and scale-invariant representations, achieving improvements of +6.0% mAP and +9.0% AP for small AAV targets on datasets such as AU-AIR and SynDrone. However, its reliance on altitude and sensor metadata limits its practical applicability. Alternatively, the interactive tiny-object detection framework [23] leverages sparse user clicks to guide multi-class detection, achieving higher accuracy on Tiny-DOTA and LCell while reducing annotation time. Although effective, its main drawback remains sensitivity to inaccurate user inputs, which can affect the reliability of the detection.

Focusing on edge-optimized inference, the architecture introduced by [24] eliminates non-maximum suppression (NMS) and incorporates enhancements such as ProgLoss, STAL, and the MuSGD optimizer, delivering up to 43% faster CPU inference without major accuracy loss, though slight performance decrements can arise from aggressive optimizations on large-scale datasets.

Finally, the feature fusion framework integrated into YOLOv10 [25] addresses challenges in AAV-based object detection, particularly for small objects in complex backgrounds. By introducing Fusion Down Sampling (FDS), Fusion Up Sampling (FUS), and Fusion Multi-Head Self-Attention (FMSA) modules, this approach enhances both localization and classification, demonstrating improvements of 2–4% mAP_{50} over baseline models on the VisDrone2019 and DOTA-v1.5 datasets.

III. METHODOLOGY

In this section, we describe the proposed method in detail. We begin with an overview of the baseline architecture, followed by a description of each proposed component and its implementation details. Detecting small objects in AAV imagery remains a challenging problem due to three principal factors: extreme scale variation across objects, insufficient preservation of fine-grained spatial detail, and limited efficacy of conventional multi-scale feature fusion strategies. To address these limitations, we propose SPAF, a modified YOLO26s-based detection framework that introduces three targeted enhancements: SVA, PBAP-Net, and ACMF-Head, as illustrated in Fig. 1. Each component is described in detail in the following subsections.

A. STRUCTURE OF YOLO26

For this study, SPAF is built upon YOLO26, the latest model in the YOLO series developed by Ultralytics [26]. Like previous versions, it consists of three main components: backbone, neck, and detection head, and comes in five versions (n, s, m, l, x), which differ in channel width, depth, and maximum number of channels. The input image is first processed by the backbone, which extracts features through multiple convolutional operations and includes the C3k2 block for higher-level feature representation. These features are passed to the neck, which performs multi-scale feature fusion using an SPPF module with shortcut connections, a C2PSA block for global context via self-attention, and upsample and concatenation operations. The detection head then predicts objects at three scales (small, medium, large). YOLO26 introduces several key improvements. First, the Distributed Focal Loss (DFL) is removed to simplify deployment and improve edge compatibility. The model is also fully end-to-end, eliminating the need for NMS and reducing inference latency. Additionally, ProgLoss and the Spatial-Temporal Adaptive Layer (STAL) enhance small-object detection, while the MuSGD optimizer improves convergence speed and stability. Overall, YOLO26 is lighter, faster, and more deployment-efficient while maintaining strong detection performance.

B. SPARSE VISION ATTENTION (SVA)

Multi-scale object detection requires attention mechanisms [27] that capture both channel-level global context and fine-grained local spatial relationships without increasing computational costs. Inspired by Vision Transformers (ViT) [28] which partition images into patches and apply Multi-Head Self-Attention (MHSA) to model long-range spatial dependencies, as well as NSSA [10], which enhances the feature representation of non-semantic features through sparse attention, we propose SVA, which applies spatial MHSA to capture positional relationships between local tokens. Furthermore, SVA introduces a channel weights branch, where globally informed channel importance weights modulate spatial features, enabling richer spatial-channel representations. It is integrated into the YOLO26 backbone and is designed to capture rich spatial-channel feature

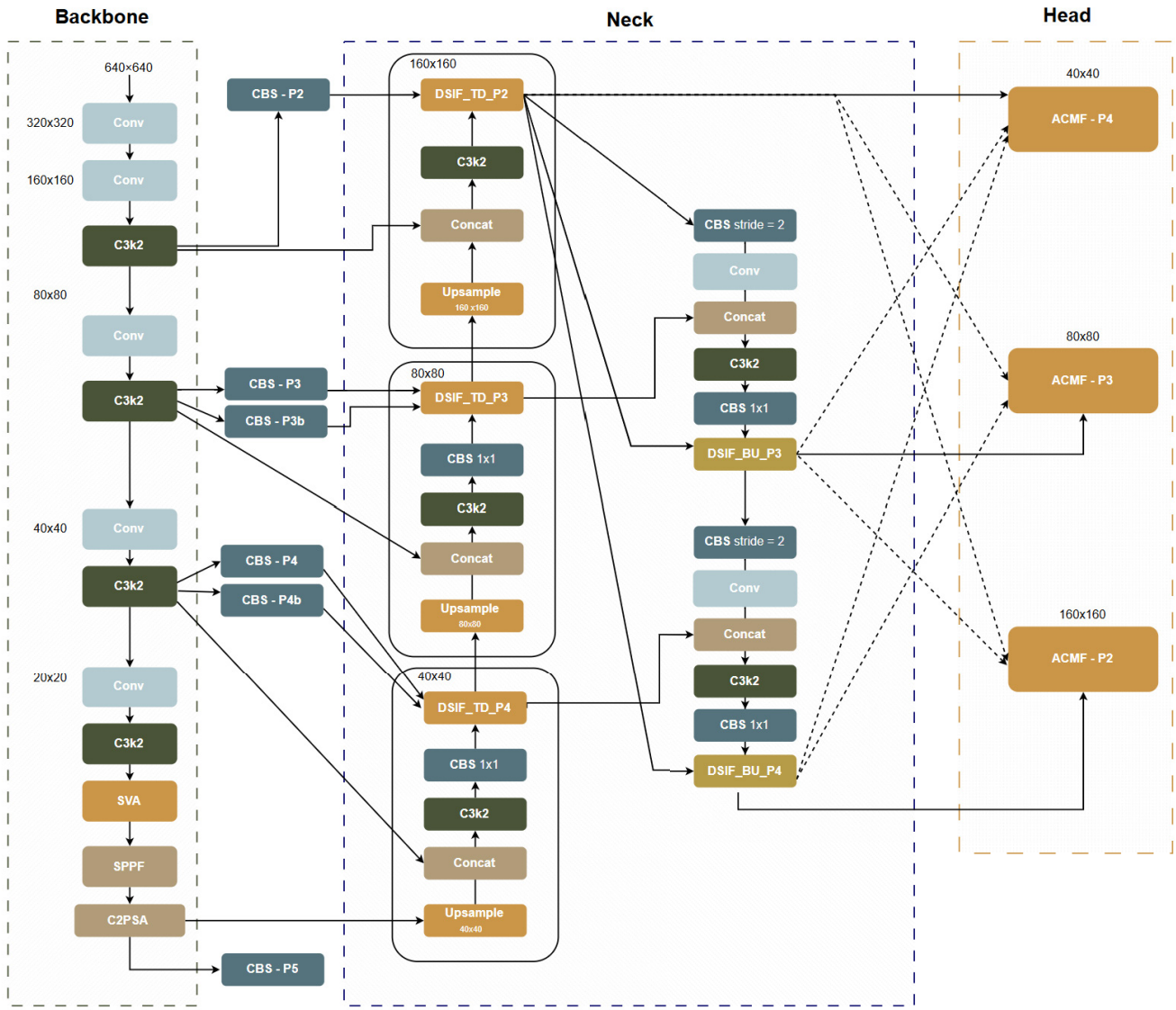


FIGURE 1. The overall structure.

representations while maintaining computational efficiency suitable for real-time detection, as shown in Fig. 2.

SVA combines two complementary branches. Given an input feature map, both branches process the input in parallel. The first branch generates a channel-wise importance weight vector to highlight informative feature channels, as shown in (1):

$$F_w = \sigma(\text{AvgPool}(\text{Conv}(X))) \tag{1}$$

where Conv is a convolutional layer, AvgPool denotes the average pooling, and σ is the sigmoid activation function that normalizes the output to [0, 1].

This weighting vector selectively enhances informative channels while suppressing redundant ones, enabling Branch 2 to focus on spatial features. The second branch applies sparse block self-attention to model local relational structure with reduced computational overhead. The input feature map X is divided into interleaved sparse blocks using a sparse coefficient S, yielding a rearranged tensor of shape $(C,\ S\times S,\ H/S\times W/S)$:

$$X_s = \text{Sparsify}(X,\ S) \tag{2}$$

The sparse coefficient S controls the balance between spatial coverage and computational efficiency by determining the number of sparse blocks. When S=1, the operation degenerates to global self-attention over the full feature map, incurring quadratic complexity $\mathcal{O}((HW)^2)$ that becomes computationally prohibitive at the 640×640 input resolution used in this work. Conversely, large values of S (e.g., S=4 or S=8) partition the feature map into excessively small blocks, severely restricting the receptive field of each attention operation and undermining the ability to capture inter-object spatial relationships critical for detecting clustered small objects in AAV imagery. At S=2,

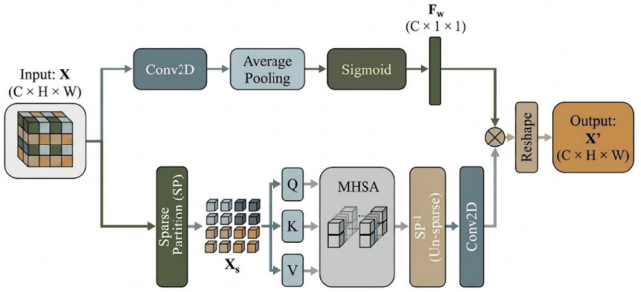


FIGURE 2. Structure of the SVA module, where X_s is the sparsified feature tensor obtained by applying sparse partitioning (SP) to the input X. The SVA has two branches: a Global Context and Channel Branch F_w, and a sparse spatial attention branch that applies MHSA within each sparse block. The outputs are fused element-wise to produce X'. The figure shows the case where $S = 2$, dividing the input into four non-overlapping blocks. When $S = 1$, the operation reduces to global attention.

each attention block covers one quarter of the spatial domain, preserving sufficient context for small object detection while reducing self-attention complexity by a factor of S^2 relative to global attention. Accordingly, S is explicitly set to 2 as a balanced design choice in this work, consistent with findings in related sparse attention literature [10]. A systematic sensitivity analysis over $S \in \{1, 2, 4, 8\}$ and attention head counts is planned as part of the ablation extensions described in Section V.

Following ViT's patch-level relational modeling, MHSA is then computed independently within each sparse block:

$$\mathrm{MHSA}(Q, K, V) = \mathrm{Concat}(\mathrm{head}_1, \mathrm{head}_2, \ldots, \mathrm{head}_h)W^O \tag{3}$$

where

$$\mathrm{head}_i = \mathrm{softmax}\left(\frac{Q_i K_i^\top}{\sqrt{d_k}}\right)V_i \tag{4}$$

where Q, K, V are the query, key, and value matrices of the sparsified block tokens, h denotes the number of attention heads, W^O represents the parameter projection matrix, and Q_i, K_i, V_i are the query, key, and value projections for the i-th head, respectively. The term d_k is the per-head dimension. By restricting self-attention to individual sparse blocks and distributed heads rather than the full feature map, computational complexity is substantially reduced compared to global self-attention.

Following attention computation, the multi-head outputs are concatenated and linearly projected via W^O. The blocks are then restored to the original spatial layout via an inverse sparsification operation $\mathrm{Sparsify}^{-1}$, and a subsequent convolution integrates the attended features and realigns the channel dimensions:

$$F_{sa} = \mathrm{Conv}\Big(\mathrm{Sparsify}^{-1}(\mathrm{MHSA}(X_s))\Big) \tag{5}$$

The outputs of both branches are fused through element-wise multiplication:

$$X' = F_w \otimes F_{sa} \tag{6}$$

where $\otimes$ denotes element-wise multiplication with broadcasting.

This fusion allows the channel importance weights from Branch 1 to selectively amplify or suppress the spatially attended features from Branch 2. Consequently, fine-grained local features extracted through sparse attention are calibrated by globally informed channel context, producing a rich feature representation that captures both local non-semantic spatial details and channel-level global dependencies.

C. PROGRESSIVE BIDIRECTIONAL AUXILIARY PYRAMID NETWORK (PBAP-NET)

The baseline detector employs a Path Aggregation Network (PANet) with Feature Pyramid Network (FPN) with dual pathways to propagate features between high-level semantic and low-level spatial representations, enabling multi-scale aggregation of information [29], [30]. Similarly, PANet- and BiFPN-based designs improve cross-scale feature interaction through bidirectional feature propagation across different pyramid levels [31]. However, this design is suboptimal for detecting small objects. First, repeated downsampling reduces spatial resolution, leading to insufficient geometric information for accurate localization. Second, long propagation paths limit the reuse of early-stage features. Additionally, conventional feature pyramids suffer from low feature reuse efficiency, where early-stage information is progressively diluted through repeated transformations. As a result, critical cues for detecting small objects become degraded. To address these limitations, PBAP-Net is proposed. It enhances feature propagation and fusion by preserving shallow features and improving cross-scale interaction efficiency. PBAP-Net enables explicit bidirectional information flow by jointly propagating features from shallow to deep layers and vice versa [10]. Accurate object localization relies on capturing fine-grained spatial information. Therefore, a second detection branch is introduced at a resolution of 160×160. This addition improves the extraction of fine-grained spatial details from early feature maps [2]. Meanwhile, the detection branch for large objects is removed to reduce computational complexity. Furthermore, six lightweight 1×1 convolutional layers are introduced as auxiliary feature carriers. These adapters preserve and forward backbone features to deeper layers, enabling direct multi-scale interaction and mitigating feature degradation caused by repeated transformations.

To further enhance feature fusion, the proposed PBAP-Net incorporates a Detail-Semantic Interaction Fusion (DSIF) module, as illustrated in Fig. 3. As a core sub-component of PBAP-Net, the DSIF module is designed to explicitly model the interaction between high-level semantic representations and low-level structural details, thereby enabling more effective cross-scale information exchange [10], [32].

Let f_i^0 denote the input feature at level i. A convolution is applied to unify the channel dimension:

$$f_i^1 \in \mathbb{R}^{H_i \times W_i \times c} \tag{7}$$

where H_i, W_i, and c represent spatial resolution and channel size, respectively.

To align features across different levels, each feature f_j^1 is resized to match the resolution of level i:

$$f_{ij}^2 = \begin{cases} D(f_j^1, (H_i, W_i)), & j < i \\ C(f_j^1), & j = i \\ U(f_j^1, (H_i, W_i)), & j > i \end{cases} \tag{8}$$

where $D(\cdot)$, $C(\cdot)$, and $U(\cdot)$ denote adaptive pooling, convolution, and bilinear interpolation, respectively.

The aligned features are concatenated:

$$F = \text{Concat}(f_{i1}^2, \ldots, f_{iN}^2) \tag{9}$$

Then, channel-aware modulation is applied:

$$\hat{F} = F \oplus (F \otimes \delta(\text{PW-Conv}(\text{GAP}(F)))) \tag{10}$$

where $\oplus$ and $\otimes$ denote element-wise addition and multiplication.

The refined features are split into branches:

$$f_{i1}^3, \ldots, f_{iN}^3 = \text{Split}(\hat{F}) \tag{11}$$

Each branch is smoothed:

$$f_{ij}^4 = \theta_{ij}(f_{ij}^3) \tag{12}$$

Finally, multi-scale interaction is enforced via element-wise multiplication:

$$f_i^5 = H([f_{i1}^4, \ldots, f_{iN}^4]) \tag{13}$$

where $H(\cdot)$ denotes the Hadamard element-wise product operation.

This design improves multi-scale feature interaction and enhances feature representation across different scales.

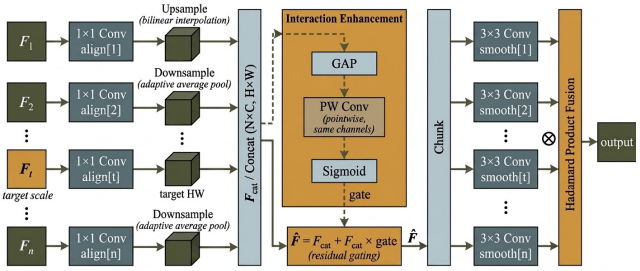


FIGURE 3. Structure of detail-semantic interaction fusion modules.

D. ADAPTIVE CHANNEL MULTI-SCALE FUSION HEAD (ACMF-HEAD)

Standard detection heads usually fuse feature maps from different pyramid levels through simple concatenation or element-wise addition, without considering the relative importance of each scale for detecting a specific object. This often leads to conflicts between features from different scales and reduces localization accuracy, especially for small objects that rely more on detailed high-resolution feature maps. To address these limitations, we propose the ACMF-Head, which dynamically learns fusion weights by jointly exploiting scale-wise and channel-wise attention. This mechanism integrates the strengths of Adaptive Spatial Feature Fusion (ASFF) [32] and Channel-Scale Feature Aggregation (CSFA) [10], as shown in Fig. 4, while introducing three key distinctions: first, a 1×1 convolution projection stage to unify channel dimensions across pyramid levels prior to any fusion operation; second, a depthwise convolution applied after spatial alignment to produce more discriminative intermediate representations for channel attention; and third, a hybrid alignment strategy employing transposed convolutions for upsampling and strided convolutions for downsampling, reducing information loss during scale transformation.

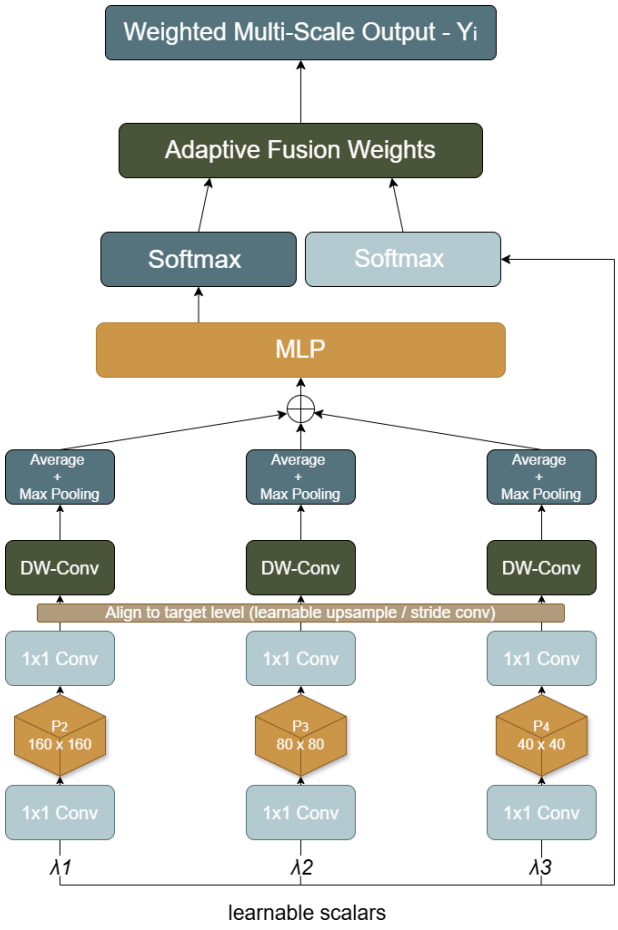


FIGURE 4. Adaptive multi-scale and channel-wise fusion mechanism in the ACMF-Head.

Given feature maps from three FPN levels P_2, P_3, and P_4, each map is first projected to a unified channel dimension C through a 1×1 convolution, resulting in projected features $\tilde{F}_i$, where $i \in \{2, 3, 4\}$. The projected features are then spatially aligned to the target level l through the hybrid alignment strategy described above, producing aligned features $F_i^{(\to l)}$. To model the relative importance of each

pyramid level, learnable parameters λ_i are introduced, and scale-wise weights $\alpha_i^s \in [0, 1]$ are computed using a softmax function.

$$\alpha_i^s = \frac{\exp(\lambda_i)}{\sum_{j=2}^{4} \exp(\lambda_j)}, \quad \text{such that} \quad \sum_{i=2}^{4} \alpha_i^s = 1. \tag{14}$$

Following spatial alignment, a depthwise convolution is applied to each aligned feature map to produce an enhanced feature representation that strengthens local spatial details prior to channel attention pooling:

$$\hat{F}_i^{(\to l)} = \text{DWConv}\left(F_i^{(\to l)}\right) \tag{15}$$

Parallel average and max pooling are then applied to $\hat{F}_i$ and fed into a shared Multi-Layer Perceptron (MLP) to produce channel-wise descriptors:

$$c_i = \text{MLP}\left(\text{AvgPool}(\hat{F}_i^{(\to l)}) + \text{MaxPool}(\hat{F}_i^{(\to l)})\right) \tag{16}$$

Applying the softmax function across $\{c_2, c_3, c_4\}$ yields the channel-wise weights α_i^c. The final fusion weights for each pyramid level are then computed by linearly combining both attention signals using learnable parameters to form an intermediate weight $\hat{\alpha}_i$

$$\hat{\alpha}_i = w_1\, \alpha_i^c + w_2\, \alpha_i^s \tag{17}$$

where w_1 and w_2 are shared across all detection levels, allowing the model to globally balance the contribution of channel-wise and scale-wise attention. To ensure that the final coefficients function as a valid convex combination summing strictly to 1 across all levels, a subsequent softmax normalization is applied:

$$\alpha_i = \frac{\exp(\hat{\alpha}_i)}{\sum_{j=2}^{4} \exp(\hat{\alpha}_j)} \tag{18}$$

The fused feature map at target level l is subsequently obtained as:

$$Y_l = \alpha_2 \cdot F_{P_2 \to l} + \alpha_3 \cdot F_{P_3 \to l} + \alpha_4 \cdot F_{P_4 \to l} \tag{19}$$

This strategy reduces conflicts between features of different scales and ensures consistent representation across detection levels. Using adaptive fusion, the model achieves improved robustness to scale variation and improved small object detection performance.

Therefore, differentiating the roles of ACMF-Head and PBAP-Net within the proposed framework is essential. PBAP-Net operates during the feature aggregation stage, focusing on enhancing bidirectional information flow and facilitating rich semantic and spatial interactions across multiple scales. In contrast, ACMF-Head is designed for the prediction stage, where it improves the final feature representations through adaptive scale- and channel-aware fusion. Consequently, the two modules serve complementary rather than redundant functions, forming a sequential refinement framework.

IV. EXPERIMENTS AND RESULTS

A. DATASETS

This study uses the VisDrone2019 dataset [33], a comprehensive benchmark for evaluating AAV-based object detection models, It is worth emphasizing that VisDrone2019 is a well-established, publicly available benchmark widely adopted by the AAV object detection community, as evidenced by all compared methods in Table 3 reporting results on the same dataset. It was collected and released by the AISKYEYE team at Tianjin University and has been used as a standard evaluation protocol in numerous peer-reviewed publications, and is therefore not a self-constructed or domain-specific dataset. It is composed of real-world images collected from 14 cities in China from numerous angles in both urban and rural environments with a variety of scene densities. The VisDrone2019 dataset has a total of 10, 209 images, which have been captured in multiple resolutions, ranging from 960 × 540 to 2000 × 1500 pixels. It is formally divided into training (6, 471 images), validation (548 images), and testing (1, 610 images). The annotated targets are categorized into 10 categories, namely: pedestrian, people, bicycle, car, van, truck, tricycle, awning-tricycle, bus, and motor, as shown in Fig. 5, where the distribution of instances across all categories in the dataset is illustrated.

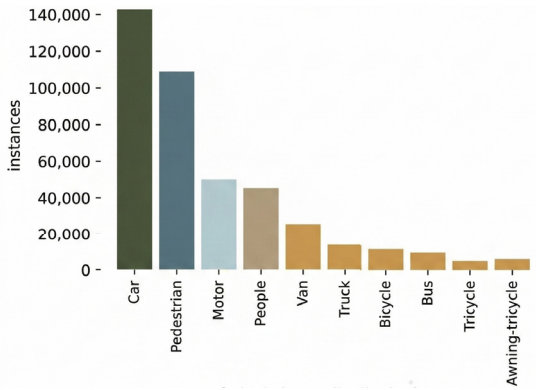


FIGURE 5. Class distribution of the VisDrone2019 dataset.

Object sizes are categorized according to the COCO standard into three sizes, where objects having a bounding box area less than 32 × 32 pixels are defined as small, between 32 × 32 and 96 × 96 pixels as medium objects, and objects exceeding 96 × 96 pixels as large. Table 1 summarizes the distribution of object sizes into categories. In addition, there are several challenges presented by the VisDrone2019 dataset, including high proportions of small objects, large scale variations, motion blur, class imbalance, high density of scenes, and frequent occlusion making it a rigorous benchmark for multi-scale AAV object detection.

B. IMPLEMENTATION DETAILS

The proposed model was implemented using the Ultralytics framework and all experiments were conducted in a local Jupyter Notebook environment based on an Anaconda

TABLE 1. Distribution of target sizes in the VisDrone2019 dataset.

Category	Small (Number)	Proportion (%)	Medium (Number)	Proportion (%)	Large (Number)	Proportion (%)
Pedestrian	65227	82.22	13804	17.40	306	0.39
People	23497	86.84	3463	12.80	99	0.37
Bicycle	7111	67.85	3244	30.95	125	1.19
Car	69862	48.22	63079	43.45	11928	8.23
Van	10751	43.08	11517	46.15	2688	10.77
Truck	3982	30.93	6725	52.24	2186	16.84
Tricycle	2176	45.22	2425	50.39	211	4.38
Awning Tricycle	1360	41.90	1699	52.39	179	5.52
Bus	1661	28.03	2365	55.10	1000	16.87
Motor	21986	74.16	7397	24.93	278	0.91
Total	207613	60.49	116616	33.98	18976	5.53

virtual environment. In terms of hardware, a workstation equipped with an NVIDIA GeForce RTX 4090 GPU with CUDA support was utilized. The deep learning environment was configured with Python 3.12.12 and PyTorch 2.10.0, along with the necessary dependency libraries to ensure the stability and reproducibility of the experiments. Our baseline model is YOLO26, upon which all proposed modules were integrated and evaluated. The main experimental settings are summarized as follows. The model was trained for 200 epochs with a batch size of 8 and an initial learning rate of 0.01. The input image resolution was fixed at 640 × 640 pixels to ensure consistency between training and inference. The weight decay was set to 0.0005 to mitigate overfitting, while the momentum parameter was set to 0.937 to accelerate convergence. In addition, MuSGD was adopted as the optimizer to enhance training stability. Furthermore, ablation experiments were conducted to evaluate the individual contribution of each module as well as their combined effect on overall performance. To ensure fairness and reproducibility, the same experimental environment and hyperparameters were consistently applied across all configurations. To ensure a rigorous and standardized benchmark for the inference speed evaluation (FPS), the performance was measured using a batch size of 1 on a sequence of 100 unseen images from the VisDrone2019-DET test-dev dataset. To eliminate initial hardware initialization overhead, a strict GPU warm-up phase was executed using the first 5 images prior to timing. The execution time was measured using native PyTorch environments under standard full-precision (FP32) without employing hardware-specific inference optimization engines (such as NVIDIA TensorRT or FP16 half-precision quantization). The reported speed of 36.7 FPS reflects the total time elapsed for the entire inference loop, demonstrating the model's inherent capacity for practical real-time deployment. Regarding statistical reproducibility, we acknowledge that reporting mean and standard deviation across multiple independent runs would strengthen the statistical grounding of individual module contributions, particularly for the SVA module which yields a +1.3% mAP@0.5 gain. Importantly, the SVA contribution is further corroborated by its consistent positive effect across all pairwise module combinations: Rows 5, 6, and 8 of Table 2 consistently show that adding SVA improves both mAP@0.5 and AP_{small} regardless of which other modules are present, providing convergent evidence of its benefit beyond a single isolated run. Multi-seed validation with mean ± std reporting is planned as part of the future work described in Section V.

C. EVALUATION METRICS

In object detection, the key metrics for evaluating performance include Precision (P), Recall (R), Average Precision (AP), Mean Average Precision (mAP), F1 Score, and Intersection over Union (IoU) [34]. These metrics quantify how successfully a model recognizes and localizes objects in video or image frames. In addition, Floating Point Operations and Giga Floating Point Operations (GFLOPs) are utilized to measure computational complexity. **P** is defined as the ratio of true positive predictions to the total number of positive predictions, while **R** is defined as the ratio of true positive predictions to the total number of actual objects. The calculation methods for P and R are respectively expressed as follows:

$$P = \frac{TP}{TP + FP} \tag{20}$$

$$R = \frac{TP}{TP + FN} \tag{21}$$

where True Positives (TP) are the number of correctly detected objects, False Positives (FP) are the number of incorrectly detected objects, and False Negatives (FN) are missed detections.

The IoU measures the overlap between the predicted bounding box and the ground-truth box. A higher IoU score indicates that the predicted bounding-box coordinates more closely align with the ground-truth coordinates. IoU is defined as follows:

$$IoU = \frac{Area(B_p \cap B_{gt})}{Area(B_p \cup B_{gt})} \tag{22}$$

where B_p is the predicted box and B_{gt} is the ground-truth box.

The AP represents the area under the Precision-Recall curve for a single class, capturing the trade-off between

precision and recall across all confidence thresholds, given as follows:

$$AP = \int_0^1 P(R)\, dR \tag{23}$$

where $P(R)$ represents the precision as a function of recall.

The mAP is the mean of AP values for all classes and serves as a comprehensive evaluation metric for multi-class object detection tasks, expressed as follows:

$$mAP = \frac{1}{N} \sum_{i=1}^{N} AP_i \tag{24}$$

where N is the total number of classes. The metric $mAP@0.5$ refers to the mean AP calculated when the IoU threshold is 0.5, while $mAP@0.5{:}0.95$ averages AP across IoU thresholds from 0.5 to 0.95. The metric AP_{small} evaluates performance on small objects with bounding-box areas less than 32×32 pixels.

The F1 Score combines P and R into a single value using the harmonic mean, providing a balanced evaluation of detection performance, especially when there is a trade-off between false positives and false negatives. A high F1 Score indicates that the model achieves a good balance between correctly detecting objects and minimizing errors, written as follows:

$$F1 = \frac{2 \cdot (P \cdot R)}{P + R} \tag{25}$$

where P denotes Precision and R denotes Recall.

The model parameter size (M) represents the total number of parameters in millions. Computational cost is measured using flops, representing the total number of floating-point operations required for a single forward pass.

The FLOPs for a convolution layer are computed as follows:

$$FLOPs = 2 \cdot H \cdot W \cdot (C_{in} \cdot K^2 + 1) \cdot C_{out} \tag{26}$$

where H and W are input feature map dimensions, K is the kernel size, and C_{in}, C_{out} are the input and output channels.

GFLOPs denote FLOPs in billions and quantify the computational cost of one full forward pass. **FPS** measures the number of frames processed per second.

D. ABLATION EXPERIMENT

To comprehensively evaluate the effectiveness of the proposed SVA, PBAP-Net and ACMF-Head modules, ablation experiments were conducted on the VisDrone2019 dataset. All experiments were performed in the same training and testing settings to ensure a fair and consistent comparison. Starting from the baseline model YOLO26s, each proposed module is introduced incrementally to analyze its individual contribution and the combined effect of all components. This setup enables a clear evaluation of the contribution of each component to overall detection performance. The results are shown in Table 2.

The experimental results demonstrate that each proposed module contributes positively to overall detection performance and that their combination yields the best results across all evaluation metrics. The SVA module leads to an improvement of 1.3% in mAP@0.5 and AP_{small} increases by 0.8%. This improvement indicates that the SVA enables the model to focus on fine-grained local features while suppressing redundant channels. The addition of PBAP-Net results in a more significant improvement, with mAP@0.5 increasing by 4.5% and AP_{small} by 3.4%, reflecting the effectiveness of its bidirectional feature propagation, the DSIF module, and the additional high-resolution 160×160 detection branch which together preserve fine-grained spatial details across pyramid levels. However, this improvement comes at the cost of a significant increase in computational complexity, with GFLOPs rising to 81.3, which is inherent to its richer feature fusion mechanism. Finally, the ACMF-Head module improves mAP@0.5 by 3.9% and AP_{small} by 2.9%, while maintaining the same parameter count as the baseline at 9.5M. This demonstrates that the adaptive consistency mechanism effectively resolves multi-scale feature conflicts. The pairwise combination experiments further validate the complementary nature of the proposed modules. Combining SVA with PBAP-Net achieves an improvement of 6.5% in mAP@0.5 and 4.7% in AP_{small}, indicating strong interaction between sparse spatial attention and bidirectional feature aggregation. Combining PBAP-Net with ACMF-Head improves by 5.1% in mAP@0.5 and 4.2% in AP_{small}, showing effective synergy between multi-scale feature enhancement and adaptive scale-channel alignment. In contrast, combining SVA with ACMF-Head results in a smaller improvement of 3.7% in mAP@0.5, suggesting that ACMF-Head benefits more when paired with PBAP-Net, which provides richer multi-scale features.

A closer look at the SVA with ACMF-Head row reveals an interesting effect. As shown in Table 2, this pairwise combination yields an mAP@0.5 of 0.430, an AP_{small} of 0.160, and an F1 score of 0.467, which is marginally lower than ACMF-Head alone (mAP@0.5 of 0.432, AP_{small} of 0.161, and F1 of 0.470) on three metrics. The reason is that ACMF-Head is designed to resolve scale conflicts among the feature maps reaching the detection head, and it works best when those features are both well-calibrated and rich in high-resolution detail. SVA alters the statistical distribution of backbone features by emphasizing fine-grained spatial-channel responses, but without PBAP-Net, the neck still relies on the standard PAN-FPN and lacks the additional 160×160 high-resolution branch. ACMF-Head therefore receives features it was not calibrated for and loses the high-resolution input that most benefits small-object reweighting, leading to a slight drop on these metrics.

When PBAP-Net is added (Row 8), this mismatch disappears. PBAP-Net supplies richer multi-scale features and the high-resolution branch, allowing ACMF-Head to exploit both the spatial detail from PBAP-Net and the refined backbone features from SVA. The full configuration reaches

0.468 mAP@0.5, 0.186 AP_{small}, and 0.505 F1, confirming that SVA and ACMF-Head are not antagonistic but require PBAP-Net as a calibration bridge between backbone-level sparse attention and head-level adaptive fusion. The three modules together produce a genuine synergistic effect that exceeds the sum of their individual gains.

Compared to the baseline model, the full proposed model achieves a **7.5%** improvement in mAP@0.5 and a **5.4%** increase in AP_{small}, along with an F1-score of 0.505, indicating a significant improvement in small object detection accuracy. These results confirm that each module addresses a specific limitation of the baseline model, and their integration produces a complementary and synergistic effect, leading to enhanced detection performance on multi-scale objects in complex AAV imagery.

E. COMPARISON EXPERIMENTS

To further validate the effectiveness of the proposed SPAF framework, we conducted comparative experiments from two perspectives. First, we compared SPAF with the baseline YOLO26s using quantitative metrics and confusion matrix analysis. Second, we evaluated SPAF against several recent state-of-the-art detectors reported on the VisDrone2019 validation benchmark.

1) SPAF vs BASELINE

The final version of SPAF achieves 0.570 precision, 0.453 recall, 0.468 mAP@0.5, 36.72 FPS, and 0.505 F1 score, improving over the YOLO26s baseline, which records 0.499 precision, 0.383 recall, 0.393 mAP@0.5, 71.02 FPS, and 0.433 F1 score. These results show consistent improvements across all evaluation metrics, demonstrating the effectiveness of the proposed architectural enhancements. The confusion matrices further explain the source of this improvement. As shown in Fig. 6 and Fig. 7, SPAF produces stronger diagonal responses across most categories, indicating more accurate classification and localization compared to the baseline. The improvement is particularly noticeable in small object dominated classes such as pedestrian and people, where correct detections increase significantly. In addition, the number of instances incorrectly assigned to the background class is reduced, suggesting that SPAF effectively minimizes missed detections. A closer inspection of the confusion matrices reveals that SPAF improves both inter-class separation and detection robustness. Classes such as motor, bicycle, and tricycle exhibit clearer boundaries, while high-frequency classes like car maintain strong and stable predictions with reduced confusion. However, some ambiguity persists between visually similar categories such as car and van or tricycle and awning tricycle, which remains a common challenge in AAV-based detection due to scale variation and occlusion. Furthermore, scale-specific evaluation using AP_{medium} and AP_{large} reveals that SPAF consistently improves detection performance across different object sizes, with more pronounced gains on medium-scale objects, highlighting its robustness to scale variation in AAV imagery Overall, these results demonstrate that SPAF provides competitive detection performance for AAV-based object detection, particularly for small and densely distributed objects, although these improvements come with increased computational cost compared with the baseline model. However, despite the increased architectural complexity, the proposed framework maintains stable training behavior due to its modular design. The impact on inference efficiency is reflected in the reported GFLOPs and FPS, indicating the computational trade-offs introduced by the additional attention mechanisms.

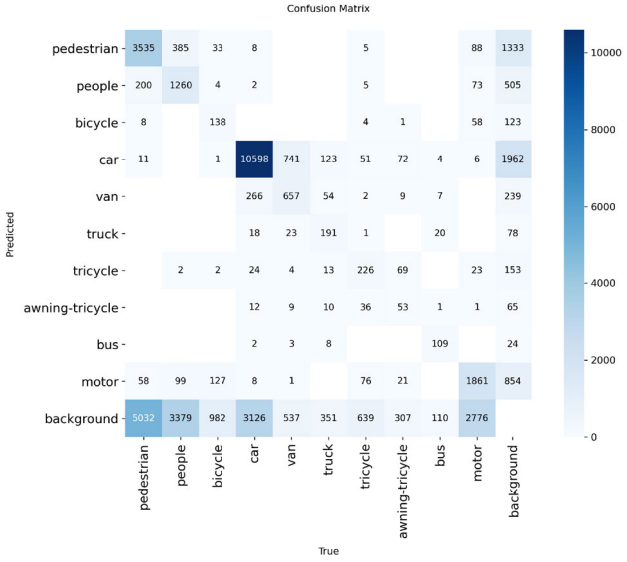


FIGURE 6. Confusion matrix of the YOLO26s.

2) SPAF vs STATE-OF-THE-ART METHODS ON VisDrone2019

We further compare SPAF with a diverse and representative set of state-of-the-art methods on the VisDrone2019 validation dataset, covering multiple research directions, including CNN-based detectors, transformer-based approaches, and AAV-oriented lightweight models. As reported in Table 3, SPAF achieves 0.468 mAP@0.5, outperforming several representative approaches such as YOLO-DroneMS [11] (0.312), MGDFIS [13] (0.394), Dynamic-Tiling [14] (0.455), and SMA-YOLO [10] (0.459), while achieving comparable performance to LMSOE-Net [12] (0.467). Lightweight-oriented entries such as LSF-YOLOv11 [35] (0.386, 12.8 M, 42.3 GFLOPs) and Model D [36] (0.376) offer strong efficiency but report lower mAP@0.5 values, reflecting the accuracy cost of their aggressive parameter reduction strategies. Although methods such as YOLOv10 [25] (0.483), AAV-DETR [20] (0.511), and CSSDet [19] (0.578) achieve higher mAP@0.5 values, they require

TABLE 2. Ablation experimental data shows that each result obtained is optimal. We use YOLO26s as the baseline model.

Model	SVA	PBAP-Net	ACMF-Head	P	R	mAP@0.5	mAP@0.5:0.95	AP_{small}	AP_{medium}	AP_{large}	GFLOPs	Params (M)	F1
1				0.499	0.383	0.393	0.232	0.132	0.303	0.408	20.5	9.5	0.433
2	✓			0.509	0.392	0.406	0.241	0.140	0.316	0.408	21.8	11	0.446
3		✓		0.537	0.420	0.438	0.264	0.166	0.335	0.410	81.3	11.5	0.471
4			✓	0.536	0.417	0.432	0.262	0.161	0.335	0.423	40.8	9.5	0.470
5	✓	✓		0.564	0.437	0.458	0.279	0.179	0.354	0.405	89.1	13	0.492
6	✓		✓	0.546	0.408	0.430	0.261	0.160	0.334	0.410	42.1	11	0.467
7		✓	✓	0.560	0.424	0.444	0.272	0.174	0.344	0.398	104.5	13.6	0.483
8	✓	✓	✓	**0.570**	**0.453**	**0.468**	**0.288**	**0.186**	**0.369**	**0.446**	**105.8**	**15.2**	**0.505**

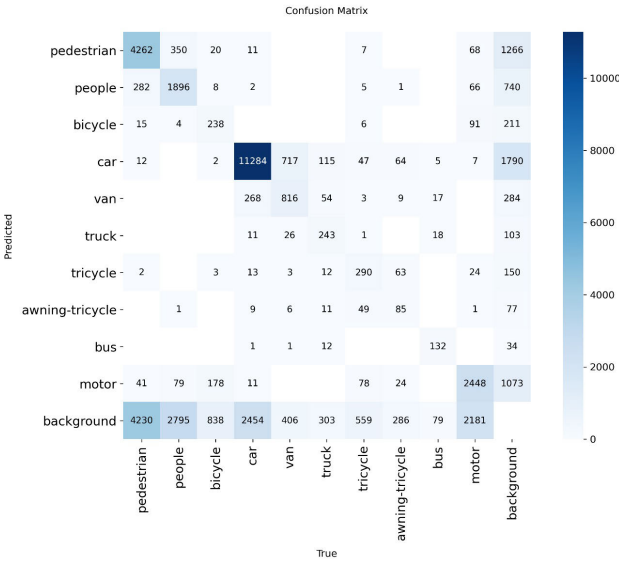


FIGURE 7. Confusion matrix of SPAF.

substantially larger model sizes and computational costs. Compared with CSSDet [19], SPAF retains approximately 81% of the reported detection accuracy while using only about 26% of its parameters and 57% of its GFLOPs. Likewise, compared with SO-DETR [15], SPAF achieves approximately 95.5% of the reported mAP@0.5 while utilizing fewer parameters. Although SO-DETR [15] achieves a higher mAP@0.5 value of 0.490, SPAF maintains competitive detection performance with a mAP@0.5 of 0.468. Similarly, compared with YOLOv10 [25], SPAF achieves nearly 97% of the reported mAP@0.5 while requiring less than half of the parameters and only around 22% of the GFLOPs. In contrast, SPAF prioritizes improved detection performance while maintaining a moderate computational cost relative to several high-capacity detectors, with 15.2 M parameters and 105.8 GFLOPs, which is considerably lower than many high-performing detectors such as YOLOv10 [25] (31.2 M, 490.9 GFLOPs) and CSSDet [19] (57.996 M, 187.0 GFLOPs). Furthermore, SPAF achieves an F1 score of 0.505, slightly surpassing SMA-YOLO [10] (0.503) and LMSOE-Net [12] (0.501), indicating a well-balanced precision and recall performance. Overall, these results demonstrate that SPAF provides a robust solution for AAV-based object detection with improved multi-scale detection capability. Compared to the baseline, it significantly reduces missed detections and improves class discrimination, particularly for small and densely distributed objects. Compared to existing methods, it achieves competitive accuracy while maintaining a practical computational cost. This makes SPAF suitable for resource-constrained AAV applications that require a balance between detection accuracy and efficiency.It should be noted that the comparative results in Table 3 are reported on the VisDrone2019 validation set, consistent with the evaluation protocol adopted by all compared methods including SMA-YOLO [10], LMSOE-Net [12], CSSDet [19], and AAV-DETR [20]. Submission to the official VisDrone2019 test-dev evaluation server is planned as part of the extended evaluation described in the future work section, to enable direct comparison under the standardized leaderboard protocol.

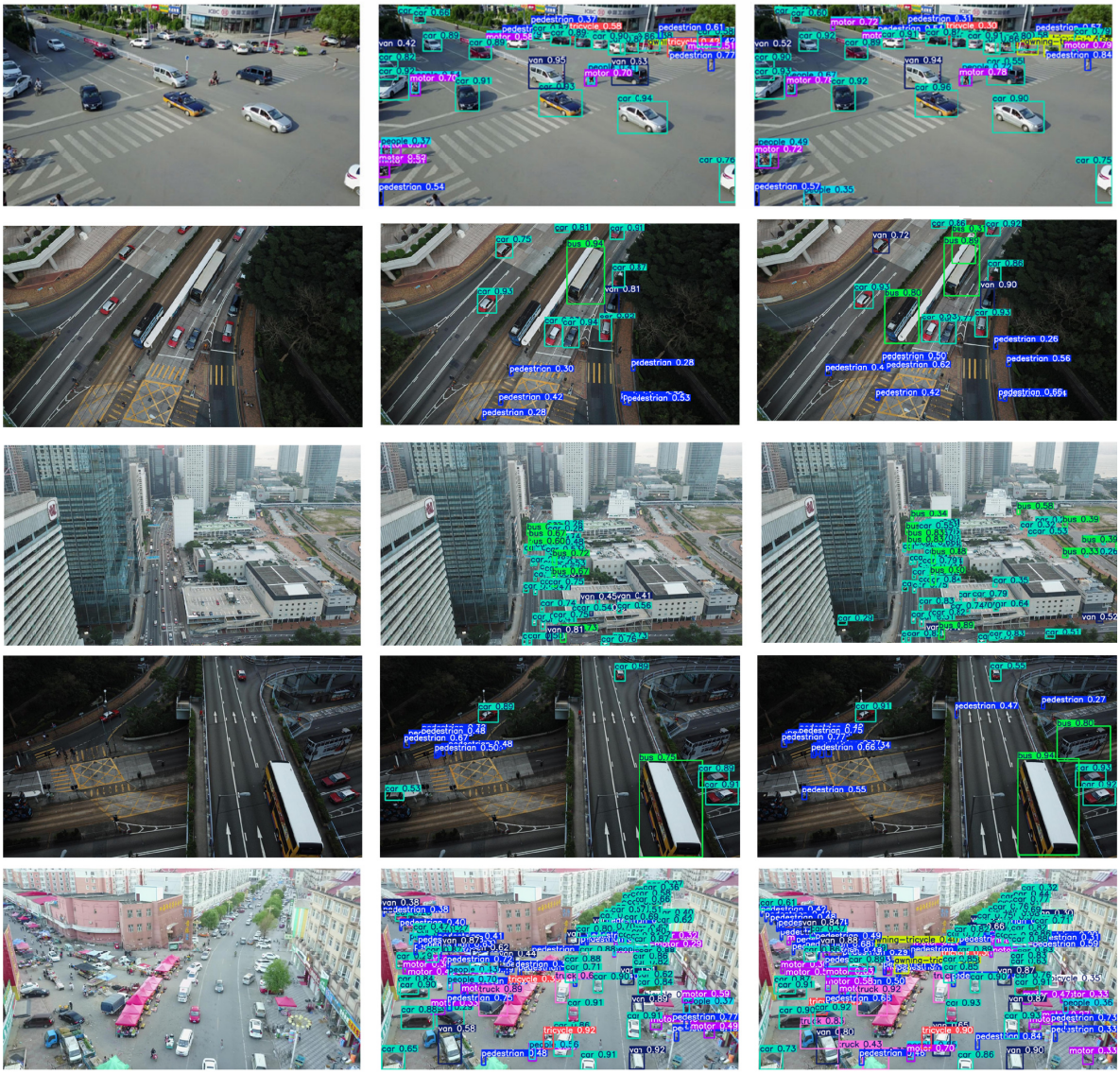
F. DISCUSSION AND VISUALIZATION

To evaluate the detection performance of the proposed model, we present qualitative comparisons with the baseline YOLO26s across five challenging scenes from VisDrone2019 dataset: high-altitude urban intersection, a low-light highway, a high-altitude cityscape, a congested highway, and a crowded street market with heavy occlusion. These scenes capture a range of difficulties including occlusion, cluttered backgrounds, low illumination, scale variation, and extreme crowding, making them suitable for evaluating multi-scale object detection, especially for small objects. The results are illustrated in Fig. 8.

As shown in Fig. 8(b), YOLO26s exhibits several limitations across challenging scenarios. In the first scene, it misses multiple pedestrians and fails to detect awning-tricycles, although it correctly identifies larger and well-separated vehicles such as vans. In the second scene, while maintaining reasonable car–van discrimination, it produces several false pedestrian detections. In the third scene, the model generates overlapping bounding boxes with inconsistent

TABLE 3. Comparison with state-of-the-art models on VisDrone2019-val. We use YOLO26s as the baseline model. The symbol "—" indicates unavailable data.

Model	P	R	mAP@0.5	Params (M)	GFLOPs	FPS	F1
YOLO-DroneMS [11]	1.00	0.52	0.312	3.14	—	—	0.35
SMA-YOLO [10]	0.573	0.447	0.459	3.3	24.9	50.1	0.503
MGDFIS [13]	—	—	0.394	—	49.9	—	—
LMSOE-Net [12]	0.568	0.451	0.467	29.1	135.7	—	0.501
YOLOv10 [25]	—	—	0.483	31.2	490.9	—	—
CSSDet [19]	—	—	0.578	57.99	187.0	14.6	—
UAV-DETR [20]	—	—	0.511	42.0	170.0	—	—
SO-DETR [15]	—	—	0.490	20.5	64.3	—	—
Dynamic-Tiling [14]	—	—	0.455	—	—	—	—
Model D [36]	—	0.483	0.376	—	—	—	—
LSF-YOLOv11 [35]	—	—	0.386	12.8	42.3	63.0	—
SPAF (ours)	0.570	0.453	0.468	15.2	105.8	36.72	0.505



(a) original image (b) baseline (Yolo26s) (c) SPAF (ours)

FIGURE 8. Visualization results comparing SPAF and YOLO26s on the VisDrone2019 dataset.

confidence scores and fails to detect objects in parts of the image. In the fourth scene, it detects only a limited number of objects, missing both buses and pedestrians despite the relatively simple background. In the fifth scene,

it struggles significantly with small and occluded objects, missing multiple pedestrians and awning-tricycles in dense regions. However, YOLO26s maintains relatively stable performance in low-density scenes, with fewer false positives and acceptable separation between similar vehicle categories.

In contrast, as shown in Fig. 8(c), SPAF achieves more consistent and accurate detections across all five scenes. In the first scene, it successfully detected previously missed pedestrians and awning-tricycles while maintaining accurate van classification. In the second scene, it corrects the pedestrian misclassifications observed in the baseline while maintaining reliable car-van discrimination. In the third scene, SPAF produces more consistent detections with reduced overlapping bounding boxes and successfully recovers objects in the regions that the baseline left entirely undetected. In the fourth scene, it detects the missed bus and substantially more pedestrians, providing more complete scene coverage. In the fifth scene, SPAF demonstrates the most significant improvement, successfully identifying small and heavily occluded pedestrians as well as awning-tricycles that the baseline completely missed, with better handling of extreme crowding and severe occlusion.

These improvements are mainly attributed to the proposed SVA, PBAP-Net, and ACMF-Head modules. SVA enhances spatial-channel feature representation through sparse attention and channel weighting, PBAP-Net improves multi-scale feature propagation via bidirectional fusion of shallow and deep features, and ACMF-Head adaptively fuses multi-scale features using channel- and scale-aware attention, reducing inter-scale conflicts and improving feature selection.

Despite these improvements, SPAF still exhibits several limitations. To provide a structured analysis of the remaining failure cases, we categorize the observed errors into three types based on the confusion matrices and qualitative visualizations in Fig. 6, Fig. 7, and Fig. 8.

First, false negatives (missed detections) occur predominantly in extremely dense and heavily occluded regions, particularly for small pedestrians and awning-tricycles, as observed in Scene 5. These cases arise because the spatial resolution, even with the additional 160×160 detection branch, remains insufficient to distinguish overlapping instances at very high object densities. Second, false positives emerge mainly from inter-class visual similarity between semantically adjacent categories such as car/van and tricycle/awning-tricycle, as reflected in the off-diagonal entries of the confusion matrices in Fig. 6 and Fig. 7.This confusion is amplified by scale variation and partial occlusion, which reduce the discriminative features available for classification. Third, localization errors manifest as overlapping bounding boxes with inconsistent confidence scores, particularly in Scene 3, where cluttered backgrounds and motion blur degrade spatial precision. Compared to the baseline YOLO26s, SPAF reduces all three error types, as evidenced by the stronger diagonal responses in Fig. 7 and the qualitative improvements in Fig. 8. Nevertheless, these remaining failure cases suggest that future work should investigate instance-aware attention and refined localization strategies for ultra-dense AAV scenes. In extremely dense regions, increased sensitivity may lead to more false positives and overlapping bounding boxes. The model may also confuse visually similar vehicle categories, particularly cars and vans, in cluttered scenes. In addition, some low-resolution and heavily occluded pedestrians are still missed. Overall, the results demonstrate that SPAF provides more robust and comprehensive detection than YOLO26s, particularly in dense and highly occluded environments.

V. CONCLUSION

This study presents an enhanced object detection framework based on the YOLO26 architecture, addressing challenges related to large-scale variations and the prevalence of small objects that often limit the performance of existing detectors. The proposed model integrates three complementary modules within a unified architecture. The SVA module improves fine-grained feature extraction in complex aerial scenes. The PBAP-Net enables effective feature aggregation from multiple layers by combining top-down semantic information with bottom-up spatial details. In addition, the ACMF-Head adaptively fuses multi-scale features and reduces inconsistencies across different levels, resulting in more accurate and reliable predictions.

Experiments conducted on the VisDrone2019 dataset demonstrate that the proposed model achieves a competitive and favorable trade-off between detection accuracy and efficiency. Specifically, compared to the baseline YOLO26, the proposed framework achieves 46.8% mAP@0.5, representing an improvement of +7.5%, while the AP_small metric is improved by +5.4%, highlighting its effectiveness in detecting small-scale objects. Furthermore, the results of the study indicate that each module contributes positively to overall performance. While maintaining a computational complexity of 105.8 GFLOPs demonstrating the effectiveness of the proposed architecture for AAV object detection.

While SPAF demonstrates strong detection performance on the VisDrone2019 benchmark, several limitations point toward meaningful directions for future investigation. First, bounding box localization and label noise. Despite the significant improvement in mAP@0.5 +7.5%, the comparatively modest gain in mAP@0.5:0.95 +5.6% reveals a remaining gap in bounding box regression accuracy, particularly for small objects. This divergence arises because the ACMF-Head's reliance on downsampled, multi-scale fused features inherently trades spatial resolution for semantic richness, which aids detection but constrains localization precision. Future work will investigate integrating a lightweight bounding box refinement sub-network into the ACMF-Head, operating on high-resolution feature maps to perform sub-pixel level corrections on preliminary box coordinates, consistent with SPAF's lightweight deployment philosophy. In addition, the impact of label noise and inaccurate annotations in VisDrone2019 remains a challenge for small-object localization. Exploring robust training strategies and

meta-learning-based noise mitigation mechanisms could improve the robustness and localization accuracy of the proposed framework.

Second, computational efficiency and broader YOLO comparisons. The full SPAF framework increases GFLOPs from 20.5 to 105.8, while FPS drops from 71 to 36.7 relative to the YOLO26s baseline. Although 36.7 FPS may be acceptable in some controlled scenarios, it falls short of the strict real-time requirements of AAV edge devices. The primary contributor is the SVA sparse attention mechanism, which applies computationally expensive branches uniformly regardless of scene complexity. To address this, future work will explore two complementary strategies: *structural re-parameterization*, which merges multi branch attention structures into a single equivalent convolution at inference time, and a *dynamic inference mechanism*, which selectively activates full attention computation only when scene complexity warrants it. Together, these strategies aim to recover inference speed without sacrificing detection accuracy. Furthermore, future work will include comparing the proposed framework against established YOLO variants, including YOLOv8s, YOLOv10s, and YOLOv11s, under identical experimental settings for a more comprehensive and standardized evaluation.

Third, ablation studies and statistical validation. Future work will conduct more comprehensive ablation studies to provide deeper insight into the proposed architectural components, including the influence of the sparse coefficient $S \in \{1, 2, 4, 8\}$, the number of attention heads, the residual depthwise convolution in the ACMF Head, and the individual contributions of the P2 detection head and the removal of the P5 head. In addition, although the current experiments were conducted under identical training configurations and evaluation protocols to ensure fair comparisons, statistical validation through multiple independent training runs was not included within the scope of the current study. Future evaluations will report the mean and standard deviation across multiple random seeds and include statistical significance analysis where appropriate, providing a more comprehensive assessment of model stability, reliability, and reproducibility.

Fourth, generalization and cross-data set evaluation. The current evaluation is restricted to the VisDrone2019 dataset. Future work will extend the evaluation to additional AAV benchmarks, such as AAVDT, DOTA-v1.5, and AI-TOD, to provide a more comprehensive assessment of cross-dataset generalization and model robustness. In addition, extending SPAF to video-based AAV detection is a promising direction, leveraging temporal dependencies across consecutive frames to improve detection consistency. Inspired by the VSLM [37] framework, future research will also investigate large scale pre-trained visual representations for AAV imagery, combined with few-shot adaptation strategies, to enable efficient transfer detection under low-resource and cross-domain scenarios. Furthermore, explainability techniques such as Grad-CAM will be explored to visualize the model attention regions and provide deeper insights into the decision-making process, enhancing the transparency and trustworthiness of the proposed framework.

Fifth, continuous learning for dynamic environments. The current SPAF model is trained on fixed datasets and cannot adapt to new scene distributions after deployment. Future work will focus on continual learning strategies, such as the selective multi-task continual learning framework (SMTC-CL), enabling progressive adaptation to dynamic environmental conditions while preserving previously acquired knowledge.

Sixth, we will primarily focus on deployment-oriented optimization, such as edge-device benchmarking, TensorRT acceleration, and memory and energy efficiency analysis for practical AAV applications.

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